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Order flow, bid–ask spread and trading density in foreign exchange markets

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ABSTRACT

This paper examines the implications of market microstructure for foreign exchange markets. We argue that the usual order flow model needs to be recast in broader terms to incorporate the transaction costs of liquidity and the limitation of price discovery through order flows that involve low trading density currencies. Using a daily data set, we find that order flows are inadequate when it comes to explaining the changes in the low trading density currencies. Alternatively, within the high trading density, both order flows and bid–ask spreads significantly affect the foreign exchange rate returns. Our findings suggest that the order flow model is better at incorporating these microstructure effects except for some currencies with a very high level of trading density.

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1. Introduction

This paper examines the implications of market microstructure for foreign exchange markets. We argue that the usual order flow model needs to be recast in broader terms to incorporate the transaction costs of liquidity and the limitations of price discovery through order flows that involve low trading density currencies. Despite the notion of liquidity having previously been applied by a large number of authors to explain the behavior of asset prices, there is no consistent conclusion as to what drives the change in the asset price. Some authors suggest that information transmission is important to price discovery in the short run (Evans and Lyons, 2002, 2008). However, others attribute the change in the asset price to the liquidity (Roll, 1984; Glosten and Milgrom, 1985; Huang and Stoll, 1997; Chordia et al., 2008).[1]

On the one hand, empirical investigations of the order flow model question the ability of the information transmission to engage in foreign exchange (FX) rate discovery through excess buying (selling) pressure.[2] That is, when orders to purchase (sell) the base currency exceed orders to sell (purchase) it, the corresponding FX rate rises (falls).[3] Several studies support the view that the order flow is to a significant degree *successful* in explaining the change in the FX rate in the short term, and possibly even in the medium term.

On the other hand, liquidity refers to the matching of buyers and sellers. It is inter-temporal in nature and is not necessarily linked to price discovery. As a simple example of this distinction, suppose that all buyers of an asset arrive on Thursday and all sellers on Friday. In this case, no trade will take place on either Thursday or Friday in the absence of counterparty. To solve this problem, it is possible to introduce the liquidity supplier who will sell to the buyers on Thursday and buy from the sellers on Friday. To provide this liquidity, a spread emerges between the ask and bid prices to compensate the middleman.

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[1] Liquidity measures of trading activities such as the bid–ask spread, trading density and components of the transaction cost are highlighted in several studies related to trading sizes, high frequency trades, market quality and adverse selection.

[2] Order flow, a measure of net buying pressure, is defined as the difference between the amount of buyer-initiated trades and the amount of seller-initiated trades. In this study, the excess buying pressure denotes the amount traded at the "paid" (ask) price minus the amount traded at the "given" (bid) price.

[3] In this study, according to the definition of the term "quote" by Electronic Broking Services (EBS), a foreign exchange (FX) rate is quoted as the local currency (denominator) per unit of the base currency (numerator). In other words, it involves quoting in fixed units of base currency against variable amounts of the local currency. For example, in Canada, a quote for U.S. dollars would be US$1 = C$1.17 and would be expressed as USD/CAD = 1.17.

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Recently, through the mediation process described in O'Hara (2003), a dispute between the two parties was settled. O'Hara (2003) clearly points out that these two concepts are related, but they are not the same for price discovery. Hence, when one wants to know what drives the change in the asset price, both information transmission and liquidity should be included as explanatory variables. As each function can influence asset prices, we first discuss how the order flow conveys the information to the FX rate markets, and then consider how liquidity enters into the determinant of FX rate.

Order flow is a major determinant of FX rates because it conveys the information through the excess buying (selling) pressure. Evans and Lyons (2002), for example, in their seminal paper report that a regression of daily exchange rate returns on daily order flow results in an *R*-square in excess of 60%. Following Evans and Lyons (2002, 2008) and Bacchetta and van Wincoop (2006) employ the macroeconomic information collection system to discuss how the indirect effect of the information transmission via the order flow affects the volatility of the FX rate.

However, any such issue is possibly misguided when the order flow *fails* to explain the low trading density currency due to the limitation of price discovery. The limitation of price discovery indicates that the price impact of the order flow should be positively correlated with the trade-size clustering. That is, the order flow of the *active* (high trading density) currency tends to be obviously larger than that of the *infrequently traded* (low trading density) currency. Evans (2011) points out that, without a great deal of trading activity, dealers will not immediately adjust their spot rate quotes via the order flow to accommodate the new information on fundamentals and the risk of providing liquidity to the market. As a result, the order flow is not suitable for investigating the behavior of FX rates when currency pairs trade with low liquidity.

The limitation of price discovery, as the measure of the market quality, is very important to the investigation of the order flow. In particular, some academic research articles argue that the explanatory power of the order flow should be positively associated with it. For example, Moulton (2005) and Gwilym and Meng (2010) highlight the relationship between the price impact of the order flow and trade-size clustering in order to explain the variations in the common liquidity measures of foreign exchange transactions. Their consistent conclusions show that the price impact of the order flow is greater when customers (traders) care more about trading precise quantities, while the number of trades and the total volumes are not significantly changed.

These challenges indicate that the order flow provides neither a model nor empirical estimates of the "liquidity based" and "clustering of transactions" price impact factor. Chan and Fong (2000) therefore assert that the volatility–volume relationship underlying stock returns could in principle be driven by either one or both components of the number of trades and the size of trades.[4] Moreover, Kaul and Sapp (2009) shed more light on the issue regarding the effect of the trading activity and dealer concentration on foreign exchange market quality. They clarify that the trading activity is an important determinant of market quality and dealer concentration.[5]

According to the above evidence, liquidity measures not only affect the volatility–volume relationship but also the information transmission ability of the order flow. We therefore suggest that the liquidity is another wedge of the microstructure that investors should look for in FX markets. Liquidity reflects the ability to buy or sell significant quantities of an asset both quickly and anonymously. To maintain liquidity, several exchange systems use liquidity suppliers, i.e., individuals who stand ready to sell or buy whenever the customer intends to buy or sell. As suppliers of liquidity, they have monopoly rights to buy at the lower *bid* price and sell at a higher *ask* price. This privilege is the primary source of compensation for offering liquidity, namely, the *bid–ask spread*.[6]

It is more important to notice that transaction data occurring in financial markets are often contaminated by market microstructure effects, such as bid–ask spreads, liquidity ratios, turnover, and asymmetric information. Market microstructure effects indicate that changes in transaction prices over very small time intervals are mainly composed of noise and carry little information about the underlying return volatility (Hasbrouck and Seppi, 2001; O'Hara, 2003). Amihud (2002) also point out that the bid–ask spread partly represents the *illiquidity premium* on the expected stock return due to the small trading size. With the same meaning, we expect that the liquidity suppliers should require more bid–ask spreads on low trading density currencies to compensate for their illiquid risks. According to these arguments, the presence of market microstructure effects is necessarily accounted for in the order flow model.

This study's first objective is to compute Stoll's realized bid–ask spread as the measure of the liquidity (Stoll, 1989). Such a spread is markedly different from the usual spread because it is easy to detect the pattern of price *reversal* and price *continuation* in a transaction. Alternatively, the usual spread is driven solely by the relative gap between the ask price and the bid price. We estimate the coefficients on the regression of the daily exchange rate return on the bid–ask spread, and order flow for each currency pair. We find that the order flow is a dominant explanatory variable for active currencies, while the bid–ask spread is a potential candidate in explaining the daily returns on infrequently traded currencies.

For the purpose of the comparison, our second objective is to provide a Generalized Method of Moments (GMMs) to understand the link between the order flow and the information dispersion, which is captured by the estimated coefficient on the variance of the order flow. Moreover, we also examine whether the variance in price changes can be attributable to the deviation in the bid–ask spread when trading density is at a high or low level.[7] We show that the contribution of information dispersion is an important determinant of the variance of exchange rate returns for active currencies. However, the empirical results for infrequently traded currencies yield the opposite conclusion because most of these currencies are insensitive to the clustering of transactions.

The remainder of this article is organized as follows. In Section 2, we introduce two common liquidity measures in economic and financial studies. In Section 3, we describe the data and the empirical procedures of the extended order flow model including the daily realized bid–ask spread. In Section 4, we present our results,

[4] For example, the order flow in large trade size categories affects the return more than in smaller size categories for NYSE stocks.

[5] According to Kaul and Sapp (2009), market quality is a key characteristic of financial markets because it summarizes the speed and accuracy with which information is impounded in prices.

[6] Several studies find that estimates of the liquidity effects are known to be sensitive to model specifications (Huang and Stoll, 1997; Chordia et al., 2008). For example, the relative spread (usual bid-ask spread) $s^{relative}_{i,t_n} = 2(p^{Ask}_{i,t_n} - p^{Bid}_{i,t_n})/(p^{Ask}_{i,t_n} + p^{Bid}_{i,t_n})$ characterizes the relative gap between bid and ask prices for the currency i at transaction n recorded in period t_n, and the larger the spread, the more illiquid the market. The Amihud (2002) statistic, $ILLIQ_{i,m} = (1/N_m)\sum_{d=1}^{N_m}|R_{i,d,m}|/V_{i,d,m}$, highlights the aggregated price impact measure based on daily returns and traded volumes, so that the larger this indicator, the lower the liquidity. $|R_{i,d,m}|$ is the absolute return and $V_{i,d,m}$ is the volume in dollars for day d in the month for the currency i. The Pastor–Stambaugh liquidity includes "non-traded" and "traded" liquidity factors, with the latter series being derived by dividing common stocks (in the CRSP monthly stocks file data) into 10 groups based on each stock's sensitivity to the "non-traded" liquidity innovation factor (as described in Pastor and Stambaugh, 2003).

[7] Several studies, especially for the covariance model of the bid-ask spread, have argued that liquidity costs can only have a second-order effect on the level of asset prices because the transactions costs are just too small relative to the equilibrium risk premium to matter (Roll, 1984; Glosten and Milgrom, 1985).

and in Section 5, we conclude. The Appendix reports the robustness checks of the GMM model.

2. Methodology

2.1. Total price change in efficient markets

According to Stoll (1989), the total price change in a foreign exchange (FX) rate may be decomposed into three components as follows:

$$\Delta V_t = \Delta F_t + \Delta R_t + \varepsilon_t, \tag{1}$$

where ΔV_t is the total price change in a currency between period $t-1$ and period t. ΔF_t is the expected price change in a currency in the absence of the bid–ask spread, which could be predicted by the order flow. ΔR_t is the price change due to the spread. ε_t is the price change due to new information; $E_t[\varepsilon_t] = 0$. Then the covariance of the total change in a currency between ΔV_t and ΔV_{t+1} can be presented in the following matrix form:

$$\Sigma = Cov\left(\begin{bmatrix}\Delta F_t\\ \Delta R_t\\ \Delta\varepsilon_t\end{bmatrix}, \begin{bmatrix}\Delta F_{t+1}\\ \Delta R_{t+1}\\ \Delta\varepsilon_{t+1}\end{bmatrix}\right) \quad \text{and} \quad \iota = \begin{bmatrix}1\\1\\1\end{bmatrix}, \tag{2}$$

where Σ is a 3 by 3 matrix. Each element of Σ is equal to the covariance of ΔX_t and ΔX_{t+1}. Therefore, $Cov(\Delta V_t, \Delta V_{t+1}) = \iota^T\Sigma\iota$.

In an efficient market, changes in price due to new information are serially uncorrelated and are uncorrelated with leading or lagged values of the price change due to the spread. This implies that:

$$Cov(\Delta V_t, \Delta V_{t+1}) = Cov(\Delta R_t, \Delta R_{t+1}). \tag{3}$$

Eq. (3) indicates that the serial covariance in the observed price change in an efficient market is due to the covariance being induced by the bid–ask spread. As a result, Roll (1984) denotes such a price change as the *price bounce* which is caused by the bid–ask spread.

2.2. Price changes due to the bid–ask spread

Under the assumption of a constant spread, the possible price changes ΔR_t, starting at the bid price, are:

$$\Delta R_t = \begin{cases} (A_t - B_{t-1}) = (1-\delta) Swithprobability \pi, \\ (B_t - B_{t-1}) = -\delta Swithprobability (1-\pi), \end{cases} \tag{4a}$$

where A_t and B_t are the ask and bid prices in period t, respectively. $(1-\delta)$ is the amount of a price *reversal* as a fraction of the spread, S and $0 \leqslant \delta \leqslant 1$. Likewise, the amount of a price *continuation* as a fraction of the spread is δ. The probability of a price reversal and continuation are given by π and $(1-\pi)$, respectively. Price reversals are assumed to be symmetric in the sense that a price increase after a transaction at the bid price has the same size as a decrease after a transaction at the ask price. Correspondingly, the probability of a price reversal is assumed to be the same regardless of whether the last transaction was at the bid or at the ask price. Price continuations are assumed to be symmetric in the same way.[8]

Under the assumption of symmetry, the possible price changes starting at the bid price can be expressed as:

$$\Delta R_t = \begin{cases} (B_t - A_{t-1}) = -(1-\delta)\ S \text{ with probability } \pi, \\ (A_t - A_{t-1}) = \delta\ S \text{ with probability } (1-\pi). \end{cases} \tag{4b}$$

Transactions in period $t-1$ are assumed to occur with equal probability at the bid or ask price. The expected price change conditional on a transaction at the bid price is:

$$E[\Delta R_t | B_{t-1}] = \pi(1-\delta)S + (1-\pi)(-\delta S) = (\pi-\delta)S.$$

Correspondingly, the expected price change conditional on a transaction at the ask price is $-(\pi-\delta)S$. Given an equal probability of a transaction at the bid and the ask price, the unconditional expected price change is $E[\Delta R_t] = (\pi-\delta)S - (\pi-\delta)S = 0$. The realized spread that the liquidity suppliers expect to earn is defined as the expected price change after they purchase less the expected price change after they sell (Stoll, 1989); that is, the realized bid–ask spread is:

$$2(\pi-\delta)S. \tag{5}$$

Note that the realized spread is the expected revenue based on two transactions: a purchase and a sale. It is evident from Eq. (4a) and (4b) that, only if $\pi = 0.5$ and $\delta = 0$, is the realized bid–ask spread in the "order processing" world and the quoted spread is the same as the realized spread (Roll, 1984). When the quoted spread is determined solely by adverse information, $\pi = 0.5$ and $\delta = 0.5$, so that the realized spread is zero (Glosten and Milgrom, 1985). When the quoted spread is determined by inventory holding costs, $\pi > 0.5$ and $\delta = 0.5$, so that the realized spread is positive but less than the quoted spread (Huang and Stoll, 1997).[9]

If it were possible to observe whether a transaction occurs at the ask price (buyer-initiated) or at the bid price (seller-initiated), the realized spread could be estimated directly, but in most cases the information is not available. Instead, the serial pattern of the price is used to estimate the spread.

2.3. Serial covariance of price change due to the spread

Under the assumption of market efficiency, the serial covariance of price changes due to the spread may be inferred from the observed price changes. The serial covariance depends on the two-period sequence of price changes.

The possible two-period sequence of price changes is assumed to start with a transaction at the bid. Assume a transaction at the bid is followed by another transaction at the bid (a continuation) or at the ask (a reversal).[10] On the basis of the transaction process

[8] Pastor and Stambaugh (2003) assert that liquidity on day t is measured by the regression coefficient on the one-lagged order flow, which is expected to be negative as it captures the return reversal. This implies that the more liquid a currency is, the smaller is the transitory price change accompanying the order flow. However, there exists the invalidation of explanation in the coefficient because a large order flow increases not only the buying pressure but also the inventory risk and transaction costs. Moreover, it is difficult to distinguish a return reversal from a price bounce by using the daily frequency. As a consequence, the coefficient on the one-lagged order flow is not an appropriate measure of the liquidity.

[9] Huang and Stoll (1997) propose that the change in the fundamental value is attributable to the percentage of the half-spread due to the adverse selection and the serially uncorrelated public information shock. However, their results show that the estimated fraction of the adverse selection is extremely small such that their presumption of the spread is violated in their basic model. The low estimates of the adverse selection cost may be partly spurious if trades bunch at the bid or ask because large trades are broken up or because buying or selling programs cause several transactions to be at an unchanged bid or ask. To solve this problem, Huang and Stoll assert that after customers sell (purchase) at the bid (ask) price, the dealer lowers (raises) the bid (ask) price relative to the fundamental price in order to increase the probability of a subsequent customer purchase (sale). Obviously, this transaction strategy of the dealer will produce a price variation such as the bid-ask bounce if the price direction of the next transaction goes against the dealer's expectations.

[10] The transaction bid-ask spread, denoted by the spread between the actual market prices of a sell order and buy order may often differ from the quoted spread. In several instances, transactions occur at prices within the bid-ask spread, perhaps because liquidity suppliers do not always update their quotes in a timely fashion, or because they intend to rebalance their own inventory at better quoted prices, or they are willing to provide discounts to customers that are trading for other reasons rather than for private information (Glosten and Milgrom, 1985).

Table 1
Summary statistics of daily currency returns, order flows, and the changes in bid–ask spreads.

	Δp_t (%)		Δx_t		Spr_t(%)		TN
	Mean	St-D	Mean	St-D	Mean	St-D	(%)
Currency pair (H)							
For the "given" (bid) price							
EUR/CHF	0.0038	0.2422	−0.0283	0.4934	0.0223	0.0760	7.907
EUR/GBP	0.0358	0.3180	−0.0136	0.1965	0.0041	0.3683	0.868
EUR/JPY	0.0058	0.6576	−0.1214	0.5064	0.0102	0.0327	15.663
EUR/USD	0.0348	0.3657	0.0448	1.2616	0.0045	0.0107	31.597
GBP/USD	−0.0019	0.4066	−0.0089	0.4312	0.0761	1.0820	2.123
USD/CAD	−0.0107	0.6059	0.0285	0.5751	0.9576	5.7703	0.627
USD/CHF	−0.0302	0.4152	0.0467	0.4055	0.0043	0.0163	11.494
USD/JPY	−0.0290	0.5847	0.0639	1.6411	0.0058	0.0125	29.262
For the "paid" (ask) price							
EUR/CHF	0.0046	0.2411	−0.0283	0.4934	0.0223	0.0760	8.023
EUR/GBP	0.0417	0.3355	−0.0136	0.1965	0.0041	0.3683	0.849
EUR/JPY	0.0057	0.6592	−0.1214	0.5064	0.0102	0.0327	15.453
EUR/USD	0.0348	0.3655	0.0448	1.2616	0.0045	0.0107	31.661
GBP/USD	0.0009	0.4056	−0.0089	0.4312	0.0761	1.0820	2.099
USD/CAD	−0.0083	0.6089	0.0285	0.5751	0.9576	5.7703	0.563
USD/CHF	−0.0300	0.4123	0.0467	0.4055	0.0043	0.0163	11.523
USD/JPY	−0.0289	0.5841	0.0639	1.6411	0.0058	0.0125	29.386
Currency pair (L)							
For the "given" (bid) price							
EUR/NOK	−0.0118	0.3708	0.0023	0.0154	0.4547	1.5663	0.027
EUR/SEK	0.0127	0.2961	0.0003	0.0337	−0.0180	0.2045	0.109
GBP/JPY	−0.0655	0.6891	−0.0238	0.3191	−0.0056	0.7099	0.125
NZD/USD	−0.0066	2.6171	−0.0002	0.0744	−0.1027	1.0411	0.074
USD/MXN	0.0121	0.3649	0.0008	0.0111	−0.0174	0.2371	0.035
USD/SGD	0.0082	0.5116	−0.0403	1.5761	0.1163	0.4747	0.016
For the "paid" (ask) price							
EUR/NOK	−0.1049	0.4730	0.0023	0.0154	0.4547	1.5663	0.027
EUR/SEK	0.0151	0.2937	0.0003	0.0337	−0.0180	0.2045	0.107
GBP/JPY	−0.0597	0.6847	−0.0238	0.3191	−0.0056	0.7099	0.102
NZD/USD	0.0275	0.9790	−0.0002	0.0744	−0.1027	1.0411	0.076
USD/MXN	0.0079	0.3788	0.0008	0.0111	−0.0174	0.2371	0.038
USD/SGD	0.0085	0.5530	−0.0403	1.5761	0.1163	0.4747	0.014

The table reports the annual means (Mean), standard deviations (St-D) of the daily FX rate return (Δp_t), the daily order flow (Δx_t) and the daily realized bid–ask spread (Spr_t) from day $t-1$ to day t for high trading density currencies (H) (the annual number of trades is larger than 50,000) and low trading density currencies (L) (the annual number of trades is between 1000 and 50,000), in which $\Delta p_t = 100\sum_i \log(FX_{t,i}/FX_{t,i-1})$ represents the cumulative hourly returns (in percentage terms) from day $t-1$ to day t, $\Delta x_t = \sum_i \Delta x_i/1000$ is computed by aggregating the hourly order flow from day $t-1$ to day t (a negative value denotes net base currency sales) and $Spr_t = 2(\hat{\pi} - \hat{\delta})S_p$ denotes the daily realized bid–ask spread from day $t-1$ to day t. *TN* denotes the proportion of an annual number of transactions for a certain currency to that of all currencies.

depicted in Eqs. (4a) and (4b), the covariance of the transaction price change is:

$$\mathrm{cov}_T \equiv Cov(\Delta R_t, \Delta R_{t+1}) = S^2[\delta^2(1-2\pi) - \pi^2(1-2\delta)]. \tag{6}$$

The details of the derivation are expressed in Appendix A. Eq. (6) is a more general version of Roll's formula. When $\pi = 0.5$ and $\delta = 0$, $\mathrm{cov}_T = -S^2/4$, which is Roll's formula. S can be reversely solved as $S = -2\{|Cov[\Delta R_t, \Delta R_{t+1}]|\}^{0.5}$.[11]

Similarly, when the spread is determined solely by adverse information, $\pi = 0.5$ and $\delta = 0.5$, so that the realized spread is zero. As noted by Glosten and Milgrom (1985), the spread decomposition is important to highlight the role of adverse information in asset returns. Another related article, Lin et al. (1995), also demonstrates that the adverse information component is negatively correlated with the trade size. Alternatively, Huang and Stoll (1997) point out that such a covariance model fails to fully decompose the adverse selection effect from the bid–ask spread. Even though it is extremely difficult for this serial covariance model to detect the adverse selection effect, it may provide a support to explore whether the price impact on the low trading density currency is attributable to the presence of adverse-selection.

According to Glosten (1987), we employ an abbreviated association between adverse-selection and exchange rate returns to show the time varying *adverse-selection* of each FX rate:[12]

$$Cov[\Delta R_t, \Delta R_{t+1}] = -\kappa S_p^2/4, \tag{7}$$

where ΔR_t, ΔR_{t+1} is the exchange rate return in period t and in period $t+1$, respectively. $S_p = (P_a - P_b)/[2(P_a + P_b)]$ denotes the bid–ask spread for the ask price P_a and the bid price P_b. $\kappa = C/(C+A)$ is the proportion of the traded spread not attributable to adverse-selection costs. A is the adverse-selection component, and C includes the inventory and order-processing components of the bid–ask spread. If κ tends to be close to zero, then the

[11] We also implement Roll's serial covariance model and obtain similar results to Huang and Stoll (1997). Our results demonstrate that the bid-ask spread has a poor performance in terms of explaining the price change when the probability and the fraction of the spread are symmetric.

[12] Glosten (1987) extends from Roll's formula of the bid-ask spread and finds that the absolute value of the covariance of price changes should be positively correlated with the adverse-selection costs.

Table 2
Estimates of daily model for high trading density currencies.

Currency	Model 1				Model 2			
	Δx_t	*Adj* R^2	Spr_t	*Adj* R^2	$\Delta(i_t - i_t^*)$	Δx_t	Spr_t	*Adj* R^2
The change in the spot exchange rate for the "given" price								
EUR/CHF	0.2915***	0.3504	−0.7248***	0.0487	−0.0421	0.289***	−0.0545	0.3480
	(6.41)		(−10.80)		(−0.65)	(5.94)	(−0.40)	
EUR/GBP	0.2049**	0.0126	−0.2064**	0.0538	−0.1221	0.1965**	−0.2105**	0.0699
	(2.20)		(−2.13)		(−1.10)	(2.26)	(−2.19)	
EUR/JPY	0.7688***	0.3479	2.1898*	0.0087	0.0284	0.7666***	1.9861*	0.3537
	(11.88)		(1.71)		(0.17)	(12.00)	(1.79)	
EUR/USD	0.200***	0.4736	2.5416	0.0024	0.0034	0.1997***	0.4871	0.4704
	(14.63)		(1.10)		(0.06)	(14.63)	(0.33)	
GBP/USD	0.0214	−0.003	−0.1185***	0.0963	−0.0549*	0.03	−0.119***	0.0931
	(0.48)		(−5.30)		(−1.81)	(0.67)	(−5.33)	
USD/CAD	0.1286**	0.0116	0.0424***	0.1608	−0.281	−0.0739	0.0453***	0.1637
	(2.58)		(9.94)		(−0.98)	(−1.14)	(23.83)	
USD/CHF	0.5347***	0.2693	−0.0318	−0.003	−0.0697	0.5362***	0.7121	0.2676
	(10.27)		(−1.13)		(−0.75)	(10.22)	(0.42)	
USD/JPY	0.2688***	0.5649	0.8173	−0.0029	−0.0833	0.2692***	0.2475	0.5628
	(14.63)		(0.23)		(−0.84)	(14.66)	(0.10)	
The change in the spot exchange rate for the "paid" price								
EUR/CHF	0.2372***	0.2339	−0.5347***	0.0254	−0.0125	0.2379***	0.0098	0.2291
	(3.39)		(−5.30)		(−0.20)	(3.18)	(0.06)	
EUR/GBP	0.0993	−0.0001	−0.0308	−0.0023	−0.0093	0.0974	−0.0309	−0.0060
	(0.92)		(0.33)		(−0.07)	(0.87)	(−0.34)	
EUR/JPY	0.7639***	0.3436	2.1223*	0.0080	0.0276	0.7619***	1.9234*	0.3488
	(11.82)		(1.69)		(0.17)	(11.92)	(1.76)	
EUR/USD	0.1996***	0.4723	2.3181	0.0014	0.0009	0.1994***	0.2728	0.4690
	(14.78)		(1.03)		(0.02)	(14.80)	(0.19)	
GBP/USD	−0.0442	−0.0011	−0.0801***	0.0425	0.3332	−0.0284	−0.076***	0.0486
	(−0.92)		(−4.03)		(1.80)	(−0.63)	(−3.70)	
USD/CAD	0.1795**	0.0258	0.0443***	0.1753	−0.239	−0.021	0.0449***	0.1733
	(5.95)		(19.03)		(−1.17)	(−1.10)	(15.74)	
USD/CHF	0.532***	0.2708	0.1788	−0.0031	−0.0742	0.5332***	0.5102	0.2690
	(10.34)		(0.09)		(−0.82)	(10.30)	(0.31)	
USD/JPY	0.2683***	0.5655	1.1051	−0.0026	−0.0706	0.2685***	0.5638	0.5633
	(14.56)		(0.32)		(−0.72)	(14.62)	(0.24)	

The table reports estimates of the daily regressions for high trading density currencies $\Delta p_t = \alpha_0 + \alpha_1 \Delta x_t + \omega_t$, $\Delta p_t = \beta_0 + \beta_1 Spr_t + \upsilon_t$ in Model 1 and $\Delta p_t = \delta_0 + \delta_1 \Delta(i_t - i_t^*) + \delta_2 \Delta x_t + \delta_3 Spr_t + \eta_t$ in Model 2 by using the Newey-West standard errors. The regressor $\Delta(i_t - i_t^*) = 100[(i_t - i_t^*) - (i_{t-1} - i_{t-1}^*)]$ is the daily change in the interest rate differential between the local country and base country from day $t-1$ to day t. Descriptions of other variables are the same as those in Table 1. The Newey-West t-statistic appears in the parentheses and *Adj* R^2 is the adjusted R-square to detect the model's goodness of fit. The asterisks

*** Significance at the 1% level.

** Significance at the 5% level.

* Significance at the 10% level.

adverse-selection cost is much larger than the inventory and order-processing costs.

Conversely, if κ tends to be close to one, then the inventory and order-processing costs are major components of the bid–ask spread. We therefore expect that κ in the case of the low trading density currency should be smaller than that for the high trading density currency, because the effect of asymmetric information tends to occur in the small-sized trading market. Here, we impose the condition $0 < \kappa < 1$ to satisfy the economic sense of positive transaction costs for A and C. Since the components of the bid–ask spread, A and C, are unobservable, we compute κ depending on the covariance of the exchange rate returns between period $t-1$ and period t under certain simplifying assumptions.[13]

3. Data description and empirical procedures

Before we introduce our FX dataset obtained from Electronic Broking Services (EBS), it may be helpful to look at the trading perspective. Osler (2009) points out that the foreign exchange market has two segments or "tiers." In the first tier, dealers trade exclusively with customers. In the second tier, dealers trade primarily with each other. In this sense, the interdealer market is the core market in which customer prices are all based on the best available interdealer prices.[14] To be more specific, based on the bid picture described by Lyons (2006), the FX dataset can be grouped into three basic types depending on the liquidity and the sizes of the sets of FX market participants as follows: (1) direct

[13] According to the Capital Asset Pricing Model (CAPM), Glosten (1987) suggests that the covariance of market returns in period $t-1$ and the true returns in period t are associated with the adverse-selection cost and bid-ask spread for each security. He also assumes that (1) true returns are independent of all past history; (2) the spread is symmetric about the true price; and (3) the gross-profit component, C including the inventory and order-processing costs of the bid-ask spread does not cause a conditional drift in prices.

[14] At each moment, any dealing bank can choose whether to supply liquidity or demand it because the dealing bank is not forced to provide the designated liquidity to the interdealer (interbank) foreign exchange market. A dealer needing liquidity can, of course, call another dealer and request a quote. Being markedly different from this market, the customer market is quote-driven, meaning that instantaneous liquidity is provided by professional market makers (Grossman and Miller, 1988). As in most such markets, currency dealers are under no formal obligation to provide liquidity, unlike specialists on the NYSE. Failing to provide liquidity on demand, however, could be costly to a dealer's reputation, and so dealers are extremely reliable.

Table 3
Estimates of daily model for low trading density currencies.

Currency	Model 1				Model 2			
	Δx_t	*Adj* R^2	Spr_t	*Adj* R^2	$\Delta(i_t - i_t^*)$	Δx_t	Spr_t	*Adj* R^2
The change in the spot exchange rate for the "given" price								
EUR/NOK	−0.4287	−0.0035	−0.0187*	0.0024	0.0335	0.5553	−0.0213	−0.004
	(−0.37)		(−1.66)		(0.63)	(0.33)	(−1.46)	
EUR/SEK	0.7685*	0.0038	−0.4374**	0.0878	0.0796	0.6385	−0.429*	0.0871
	(1.73)		(−1.97)		(0.56)	(1.48)	(−1.95)	
GBP/JPY	0.1244**	−0.0005	−0.2079***	0.0422	0.0541	0.1107	−0.2072***	0.0375
	(2.00)		(−4.78)		(0.23)	(1.65)	(−4.62)	
NZD/USD	1.6632*	−0.0016	0.1069**	−0.0021	0.3498	1.4475	0.0898**	−0.008
	(1.73)		(2.25)		(0.32)	(1.37)	(2.05)	
USD/MXN	−2.005	−0.0001	−0.094	−0.0002	0.0349	−2.1683	−0.1012	−0.003
	(−0.88)		(−0.71)		(0.14)	(−0.94)	(−0.78)	
USD/SGD	0.0723	0.0459	0.3879***	0.1260	0.2145	0.0517	0.3637***	0.1544
	(2.48)		(3.66)		(1.28)	(1.39)	(3.73)	
The change in the spot exchange rate for the "paid" price								
EUR/NOK	−0.122	−0.003	−0.1938***	0.4098	−0.0933	−0.0376	−0.1938**	0.4076
	(−0.08)		(−30.08)		(−1.44)	(−0.02)	(−20.23)*	
EUR/SEK	0.1759	−0.0035	−0.1951	0.0147	−0.2478	0.1147	−0.2046	0.0167
	(0.41)		(−1.42)		(−1.74)	(0.27)	(−1.45)	
GBP/JPY	0.1285**	−0.0003	−0.1123***	0.0097	0.0709	0.1214**	−0.1116***	0.0054
	(2.06)		(−3.40)		(0.26)	(2.02)	(−3.31)	
NZD/USD	1.0367	0.0024	0.1772***	0.0319	0.3069	0.5791	0.1724***	0.0275
	(1.39)		(3.69)		(0.68)	(0.65)	(3.51)	
USD/MXN	−2.6548	0.0023	0.4954***	0.0912	0.0766	−1.8872	0.4897***	0.088
	(−1.30)		(2.86)		(0.49)	(−1.12)	(2.84)	
USD/SGD	0.0317	0.0044	0.6625***	0.3212	−0.0866	−0.0004	0.6611***	0.317
	(0.48)		(6.45)		(−0.87)	(−0.01)	(5.95)	

The table reports the Newey-West estimates of the daily model for low trading density currencies. See the footnote in Table 2 to understand the definitions of these variables. The asterisks.
*** Significance at the 1% level.
** Significance at the 5% level.
* Significance at the 10% level.

interdealer trades, (2) brokered interdealer trades and (3) customer-dealer trades.[15]

In contrast to interdealer trades, data on customer-dealer trades have not been made available until recently by the best, unique proprietary and largest FX dataset managed by the Union Bank of Switzerland (UBS).[16] It is this dataset that has been used in Cerrato et al. (2011). However, such customer-dealer data are very difficult to obtain, because banks have no regulatory obligation to disclose such information. As a result, banks consider their trades with customers as being highly confidential. Cerrato et al. (2011) also point out that the customer-dealer trades are superior to the interdealer trades because the customer-dealer trades are fully revealed to the market.

During the past decade, the FX market has begun to shift from a phone-based network to an electronic trading platform. In 2001, less than 40% of interbank dealings were transacted through EBS and Reuters Dealing 2000. In 2007, between 75% and 90% of all interbank dealings were channeled through these two systems. Furthermore, the share of transactions involving crucial exchange rates (EUR/USD, USD/JPY, EUR/JPY, USD/CHF and EUR/CHF) dealt with through EBS is higher than that handled through Reuters.[17] For example, the daily volume of EBS was 160 billion USD for EBS compared to 105 billion USD for Reuters in 2007.

As Barker (2007) notes, these electronic portals provide a live price stream that aggregates all bids and offers posted on the system. This interbank price is continuously displayed to all market participants. However, as Osler (2009) mentions, contrary to most other limit-order markets, the FX market has low pre- and post-trade transparency.[18] In fact, pre-trade information is limited to the best bid and ask quotes and post-trade information is a listing of transaction prices while trade volumes are not published. As a result, the pre-trading (quoted) order flow is unable to capture the actual changes in the FX rate. Alternatively, we use the post-trading FX data rather than the quote data.

We consider the following two-step process to compute the daily change in the FX rate. First, to keep away from the

[15] Data on direct interdealer trades are the most liquid part of the market, but also the smallest. This is because data covering the direct interdealer trades in the inner ring of the bid picture—were limited until recently to the trades of individual dealers (versus summarizing the trades of all dealers, or some large subset). Dealers trade directly with one another using a bilateral electronic trading system which is most notably referred to as Reuters Dealing 2000-1. Relatively speaking, the middle ring contains brokered interdealer trading. The newfound accessibility of this data category corresponds to the shift in brokered trading from voice-based electronic systems or online trading systems. The main electronic brokers in the major spot markets are EBS and Reuters Dealing 2000-2. As the electronic brokerages took over, their best posted bid and offer quotes became the benchmark for market prices. By the end of the 1990s, voice brokers were only important in the "exotic" (relatively illiquid) currencies for which electronic brokers are unavailable. At the present time, as discussed in Barker (2007), direct dealing among major banks has all but disappeared. The speed of this transition reflects the intensity of competition in this market. Finally, the outer ring represents customer-dealer trading. Data from this ring come directly from banks' own order-flow records.

[16] UBS is a Swiss global financial services company headquartered in Basel and Zurich, Switzerland, which provides investment banking, asset management, and wealth management services for private, corporate, and institutional clients worldwide, as well as retail clients in Switzerland.

[17] The currencies used in this study include the Canadian dollar (CAD), the Swiss franc (CHF), the Euro dollar (EUR), the British pound (GBP), the Japanese yen (JPY), the Mexican peso (MXN), the Norwegian krone (NOK), the New Zealand dollar (NZD), the Swedish krona (SEK), the Singapore dollar (SGD) and the United States dollar (USD). The International Organization for Standardization code (ISO code) for each currency is shown in the parentheses.

[18] To be precise, there are quoted and hit types of orders (market orders) possible on the EBS. Quotes enter the order book until they reach a counterpart. Hits are orders that are directly fulfilled if a counterpart is standing in the order book, or else are immediately cancelled out otherwise.

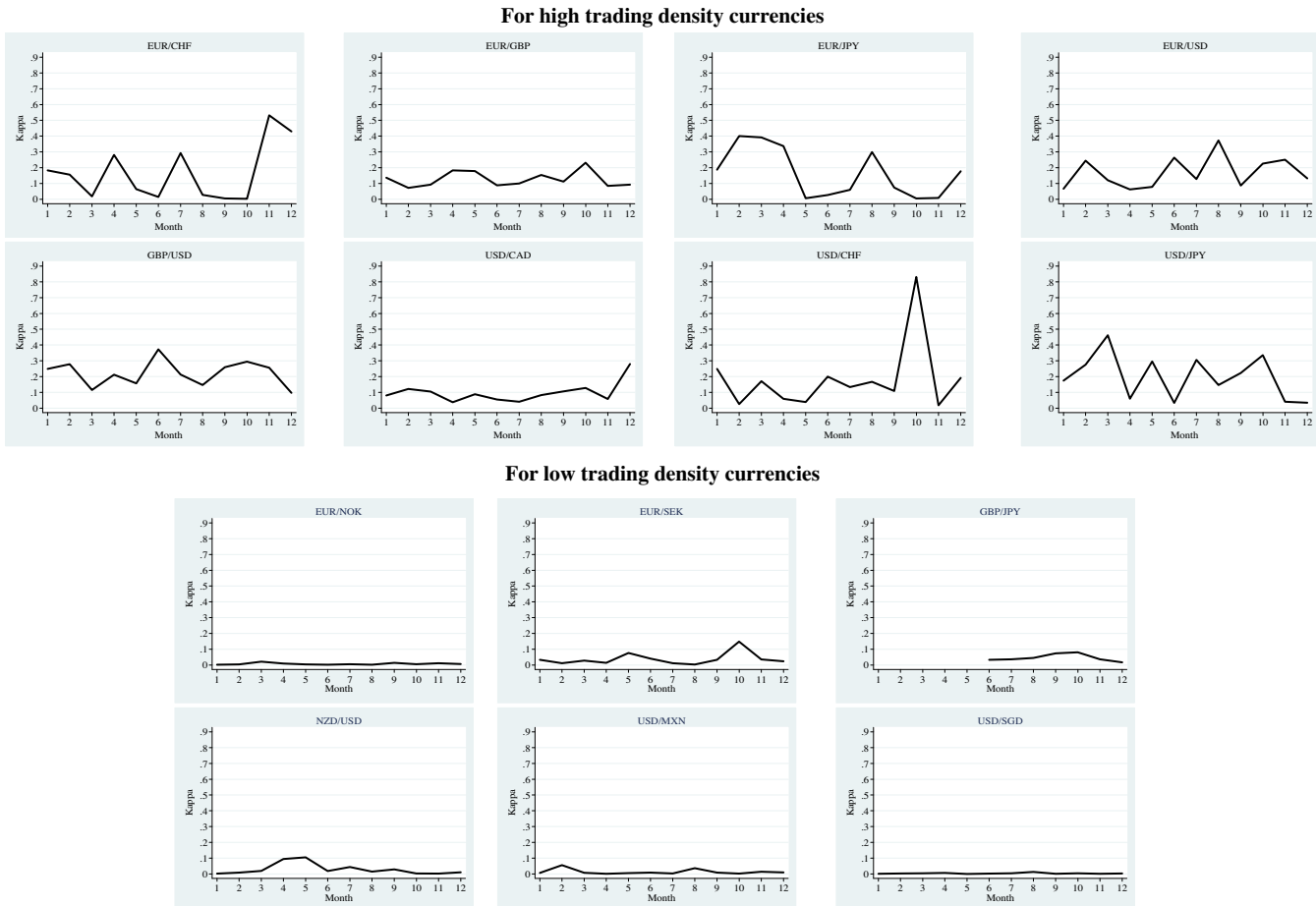

Fig. 1. Time series of the non-adverse selection components.

microstructure effect on the FX rate due to the high frequency, we take the average of the intra-hour FX rate of each currency pair to obtain the hourly post-trading FX rate. Second, we compute the changes in the post-trading FX rate for each hour and aggregate them to obtain the daily change in the FX rate.

Similarly, we follow the same procedure to obtain the daily realized bid–ask spread. We also use the hourly post-trading FX rate to compute the hourly usual bid–ask spread, which is $S_p = (P_a - P_b)/[2(P_a + P_b)]$, where P_a is the hourly FX rate for the ask (paid) price and P_b is the hourly FX rate for the bid (given) price. To obtain Stoll's daily realized spread, we estimate the amount of a price *continuation* as a fraction of the spread, δ, and the probability of a price reversal, π by averaging the total hourly δ and π in our sample period. As a result, we obtain Stoll's daily realized spread by using Eq. (5), $2(\pi - \delta)S$, where S is the same as the hourly usual bid–ask spread S_p.[19]

The hourly trading amount differs from the changes in the FX rate and bid–ask spread. We firstly aggregate the intra-hour trading amount to obtain the hourly trading amount and aggregate it again to obtain the daily amount of seller-initiated trades and the daily amount of buyer-initiated trades. We then compute the daily order flow by taking the difference between the daily amount of seller-initiated trades and the daily amount of buyer-initiated trades.[20]

We use the hourly post-trading spot FX rate and trading amount data sets in the EBS to compute the order flows and bid–ask spreads for 14 currency pairs.[21] These "deal record" sets contain most existing inter-dealer transactions and cover the sample period from Jan. 3rd, 2007 through Jan. 3rd, 2008. For the purposes of comparison with Evans and Lyons' (2002) order flow model, we transform hourly data into daily data and supplement these with daily interest rate differential data collected from Datastream.

Table 1 presents some summary statistics of daily currency returns, order flows, and the bid–ask spreads at a "given" (bid) price and "paid" (ask) price for 14 currency pairs. Of these, the daily FX rate returns, $\Delta p_t = 100 \sum_i \log(FX_{t,i}/FX_{t-1,i})$, comprise the cumulative hourly returns (in percentage terms) from day $t-1$ to day t. The daily order flow $\Delta x_t = \sum_i \Delta x_i/1000$ is computed by aggregating

[19] The daily realized bid-ask spread is derived from Eq. (5), that is, $2(\pi - \delta)S$. The usual realized spread is widely used in finance, that is, $S_p = (P_a - P_b)/[2(P_a + P_b)]$. In our study, S is the same as the usual realized spread. However, there is a difference between them. To compute the daily realized bid-ask spread, we employ not only the usual realized spread, but also the amount of a price *continuation* as a fraction of the spread, δ and the probability of a price reversal, π.

[20] We label the amount of the trades for each currency pair in EBS as the "trading amount". As conveyed by the words, "trading amount" is the total amount of each trade. In EBS, the trading amount is divided into bid (given) and ask (paid) amounts based on the trading category-side. If the side equals zero, then the trading amount is the amount of the buyer-initiated trade; if the side equals one, then the trading amount is the amount of the seller-initiated trade.

[21] Dating back to Hasbrouck's (1993) seminal paper, it is a well accepted fact that transaction data occurring in financial markets are often contaminated by market microstructure effects, such as bid-ask spreads, liquidity ratios, turnover, and asymmetric information. This is particularly relevant when dealing with high frequency data, which are often used to compute the realized volatility because changes in transaction prices over very small time intervals are mainly composed of noise and carry little information about the underlying return volatility (see also, e.g., Hasbrouck and Seppi, 2001 and O'Hara, 2003).

Table 4
Generalized Method of Moments (GMM) estimates of daily high trading density currencies.

Currency	γ_1	γ_2	$E[Var_t(\Delta p_t)]$	$E[Var_t(\Delta x_t)]$	$E[Var_t(Spr_t)]$	Wald tests		
						$\gamma_1^2 = 0$	$\gamma_2^2 = 0$	$\gamma_1\gamma_2 = 0$
The change in the log spot exchange rate for the "given" price								
EUR/CHF	0.3194***	0.1632	0.0515***	0.1222***	0.0004***	17.61***	0.17	0.73
	(8.39)	(0.83)	(13.16)	(6.71)	(4.71)	[0.000]	[0.677]	[0.393]
EUR/GBP	0.1456	−0.289***	0.1005***	0.0048**	0.0929***	0.69	4.46**	2.71*
	(1.66)	(−4.23)	(8.76)	(2.41)	(4.59)	[0.406]	[0.035]	[0.099]
EUR/JPY	0.7245***	2.168**	0.319***	0.2634***	0.0007***	36.43***	1.32	4.86**
	(12.07)	(2.30)	(12.47)	(7.08)	(3.92)	[0.000]	[0.231]	[0.027]
EUR/USD	0.1952***	1.8688**	0.1232***	2.5317***	0.0001***	64.12***	0.64	2.55
	(16.02)	(1.60)	(10.68)	(5.83)	(3.10)	[0.000]	[0.238]	[0.110]
GBP/USD	0.0073	−0.115***	0.1479***	0.0208***	2.6624**	0.01	5.57**	0.02
	(0.16)	(−4.72)	(5.59)	(4.49)	(4.50)	[0.937]	[0.018]	[0.875]
USD/CAD	−0.074	0.0437***	0.3888***	0.0101***	0.7376	0.19	63.61***	0.75
	(−0.87)	(15.95)	(6.48)	(3.55)	(1.46)	[0.662]	[0.000]	[0.387]
USD/CHF	0.5067***	−0.6364	0.0895***	0.0738**	0.0002***	21.86***	0.06	0.24
	(9.35)	(−0.49)	(3.08)	(2.60)	(9.24)	[0.000]	[0.806]	[0.624]
USD/JPY	0.2652***	−0.1341	0.2392***	2.2976***	0.0002***	60.96***	0.00	0.01
	(15.62)	(−0.08)	(7.37)	(5.46)	(4.71)	[0.000]	[0.969]	[0.938]
The change in the log spot exchange rate for the "paid" price								
EUR/CHF	0.2845***	0.1944	0.0515***	0.1213***	0.0004**	6.88***	0.10	0.40
	(5.25)	(0.62)	(13.84)	(6.58)	(4.67)	[0.008]	[0.757]	[0.527]
EUR/GBP	0.0408	−0.0505	0.1001***	0.0047***	0.0813***	0.04	0.05	0.13
	(0.39)	(−0.46)	(8.62)	(2.34)	(5.03)	[0.845]	[0.817]	[0.720]
EUR/JPY	0.7212***	2.0923**	0.3209***	0.2629***	0.0007***	36.08***	1.28	4.71**
	(12.01)	(2.26)	(12.49)	(7.05)	(3.94)	[0.000]	[0.259]	[0.029]
EUR/USD	0.1948***	1.6022	0.1229***	2.5397***	0.0001***	64.24***	0.49	1.94
	(16.03)	(1.40)	(10.68)	(5.83)	(3.10)	[0.000]	[0.484]	[0.163]
GBP/USD	−0.0061	−0.082***	0.1485***	0.0185***	2.7133***	0.00	2.50	0.02
	(−0.14)	(−3.16)	(5.65)	(4.38)	(4.38)	[0.945]	[0.114]	[0.891]
USD/CAD	−0.0214	0.0443***	0.4344***	0.01***	0.6649	0.21	32.82***	0.73
	(−0.92)	(11.46)	(5.76)	(3.45)	(1.13)	[0.647]	[0.000]	[0.393]
USD/CHF	0.5043***	−0.8102	0.0924***	0.074***	0.0002***	22.14***	0.10	0.40
	(9.41)	(−0.64)	(3.37)	(2.61)	(9.19)	[0.000]	[0.751]	[0.526]
USD/JPY	0.2647***	0.1151	0.2409***	2.2935***	0.0002***	61.13***	0.00	0.00
	(15.64)	(0.07)	(7.59)	(5.45)	(4.69)	[0.000]	[0.972]	[0.945]

The table reports the GMM parameter estimates with the asymptotic *t*-statistics in parentheses. The columns labeled "Wald tests" show the test results for the coefficient restrictions listed at the top with asymptotic *P*-values reported in square brackets based on the moment conditions in Eqs. (10)–(14). The instrument vector includes the interest rate differential. We apply Wald tests to investigate the association of variances and covariance between the changes in the exchange rate, order flow and bid–ask spread. The asterisks.
*** Significance at the 1% level.
** Significance at the 5% level.
* Significance at the 10% level.

the hourly order flow from day $t-1$ to day t (a negative sign denotes the net base currency sales). $Spr_t = 2(\hat{\pi} - \hat{\delta})S_p$ denotes the daily realized bid–ask spread from day $t-1$ to day t and $S_p = (P_a - P_b)/[2(P_a + P_b)]$ is the usual realized spread. The trading density, *TN*, denotes the proportion of an annual number of transactions for a certain currency to that for all currencies. A significant feature of Table 1 is that the aggregate trading densities of the EUR/USD and USD/JPY are about 60%. As a result, the investigation of the trading density in EBS indicates that these two currencies have a large trading position.

3.1. The extended order flow model

We start to divide the full sample into high and low trading density currencies and then combine the order flow with the bid–ask spread in the regression because information-based microstructure models should incorporate the microstructure effects (O'Hara, 2003). The first step is to implement the regression of the daily exchange rate return on the order flow (the realized spread) to investigate how the magnitude of the information transmission (liquidity risk) affects the FX rate returns (Model 1):

$$\Delta p_t = \alpha_0 + \alpha_1 \Delta x_t + \omega_t; \Delta p_t = \beta_0 + \beta_1 Spr_t + \upsilon_t. \tag{8}$$

The next step involves the extended order flow model where we implement the following regression (Model 2):

$$\Delta p_t = \delta_0 + \delta_1 \Delta(i_t - i_t^*) + \delta_2 \Delta x_t + \delta_3 Spr_t + \eta_t, \tag{9}$$

where $\Delta(i_t - i_t^*) = 100[(i_t - i_t^*) - (i_{t-1} - i_{t-1}^*)]$ is the change in the interest rate differential between the local country's interest rate i_t and the base country's interest rate i_t^*.

3.2. Volatility of the foreign exchange rate

Although the linear least squares approach can help us to investigate the average effect of the excess buying (selling) pressure and the liquidity risk in the FX market, several empirical studies have argued that the volatility effects of the order flow and spread should have impacts on the FX rate returns (Evans and Lyons, 2008; Huang and Stoll, 1997). To capture such volatility effects, we implement the GMM procedures that account for the variance of the order flow and variance of the bid–ask spread on the volatility of the exchange rate returns. The GMM estimates of the parameters are derived from the following set of moment conditions:

$$E[(\Delta p_t - \gamma_1 \Delta x_t - \gamma_2 Spr_t)\Delta x_t] = 0, \tag{10}$$
$$E[(\Delta p_t - \gamma_1 \Delta x_t - \gamma_2 Spr_t)Spr_t] = 0, \tag{11}$$
$$E[\Psi_t(\Delta p_t) - Var_t(\Delta p_t) \otimes Z_t] = 0, \tag{12}$$
$$E[\Psi_t(\Delta x_t) - Var_t(\Delta x_t) \otimes Z_t] = 0, \tag{13}$$
$$E[\Psi_t(Spr_t) - Var_t(Spr_t) \otimes Z_t] = 0, \tag{14}$$

Table 5
Generalized Method of Moments (GMM) estimates of daily low trading density currencies.

Currency	γ_1	γ_2	$E[Var_t(\Delta p_t)]$	$E[Var_t(\Delta x_t)]$	$E[Var_t(Spr_t)]$	Wald tests		
						$\gamma_1^2 = 0$	$\gamma_2^2 = 0$	$\gamma_1\gamma_2 = 0$
The change in the log spot exchange rate for the "given" price								
EUR/NOK	−0.5573	−0.0111	0.0001***	0.0001	1.8201	0.03	0.16	0.18
	(−0.36)	(−0.80)	(2.77)	(0.46)	(0.05)	[0.856]	[0.687]	[0.669]
EUR/SEK	0.7533**	−0.567***	0.0802***	0.0004***	0.0458***	1.03	1.75	3.17*
	(2.03)	(−2.65)	(12.39)	(2.91)	(4.79)	[0.309]	[0.186]	[0.075]
GBP/JPY	0.1492**	−0.227***	0.7624***	0.0004***	0.3808***	1.27	9.22***	3.79*
	(2.26)	(−6.07)	(6.11)	(4.06)	(3.20)	[0.259]	[0.002]	[0.051]
NZD/USD	1.2537*	0.0553	1.1684***	0.0001***	1.4959*	0.73	0.26	1.20
	(1.71)	(1.02)	(4.17)	(5.53)	(1.95)	[0.393]	[0.609]	[0.273]
USD/MXN	−2.3444	−0.0172	0.0963***	0.0001***	0.0278***	0.32	0.01	0.03
	(−1.12)	(−0.17)	(4.58)	(3.28)	(5.94)	[0.574]	[0.934]	[0.869]
USD/SGD	0.0208	0.3992***	0.0921	0.0001***	0.1531***	0.08	3.21*	0.31
	(0.58)	(3.58)	(3.96)	(4.05)	(3.02)	[0.773]	[0.073]	[0.575]
The change in the log spot exchange rate for the "paid" price								
EUR/NOK	−0.6555	−0.196***	0.2208***	−0.0004	1.8212***	0.04	164.2***	0.17
	(−0.41)	(−25.63)	(4.96)	(−0.98)	(2.86)	[0.838]	[0.000]	[0.679]
EUR/SEK	0.2583	−0.346***	0.1004***	0.0005***	0.0429***	0.10	4.29**	0.38
	(0.62)	(−4.14)	(11.97)	(3.45)	(4.40)	[0.755]	[0.038]	[0.538]
GBP/JPY	0.1412**	−0.0984**	0.8803***	0.0004***	0.3916***	1.54	1.53	2.81*
	(2.48)	(−2.47)	(11.89)	(5.03)	(3.78)	[0.215]	[0.216]	[0.094]
NZD/USD	0.4109	0.197**	0.7118***	0.0001***	1.2936	0.06	1.53	0.26
	(0.48)	(2.47)	(4.45)	(5.84)	(1.89)	[0.812]	[0.216]	[0.61]
USD/MXN	−2.2355	0.5054***	0.1038***	0.0001***	0.0274***	0.34	4.06**	1.23
	(−1.17)	(4.03)	(8.15)	(2.85)	(6.25)	[0.559]	[0.044]	[0.267]
USD/SGD	−0.004	0.617***	0.0601***	0.0001***	0.1305*	0.00	9.71***	0.01
	(−0.11)	(6.23)	(4.16)	(4.13)	(2.56)	[0.955]	[0.002]	[0.912]

See the footnote in Table 4..
*** Significance at the 1% level.
** Significance at the 5% level.
* Significance at the 10% level.

Table A
Derivation of serial covariance of transaction price changes.

	ΔR_t			
	Initial trade at bid		Initial trade at ask	
	$-\delta S$ (BB)	$(1-\delta)S$ (BA)	δS (AA)	$-(1-\delta)S$ (AB)
ΔR_{t+1}				
$-\delta S$ (BB)	$(1-\pi)^2$	0	0	$\pi(1-\pi)$
$(1-\delta)S$ (BA)	$\pi(1-\pi)$	0	0	π^2
δS (AA)	0	$\pi(1-\pi)$	$(1-\pi)^2$	0
$-(1-\delta)S$ (AB)	0	π^2	$\pi(1-\pi)$	0

where Z_t is a vector of instruments including the change in the interest rate differential (the public information shocks) across the base and local countries.[22] Eq. (10) follows from the orthogonal condition between the shocks to prices and shocks to the order flow. Likewise, Eq. (11) presents the same meaning for the change in the bid–ask spread. Eqs. (12)–(14) combine the (un-centered) second moments of price changes, the order flow and the change in the spread implied by the model with measures of the variance of price changes, $\Psi_t(\Delta p_t)$, the variance of order flow, $\Psi_t(\Delta x_t)$ and the variance of the bid–ask spread, $\Psi_t(Spr_t)$. These measures are computed for each day in our sample from the intraday (hourly) observations as:

$$\Psi_t(\Delta p_t) = \sum_{i=1}^{t_i} \Delta p_i^2, \Psi_t(\Delta x_t) = \sum_{i=1}^{t_i} \Delta x_i^2, \quad \text{and} \quad \Psi_t(Spr_t) = Spr_t^2, \tag{15}$$

where the subscript i denotes the ith intraday observation, and t_i denotes the total number of the sequence of trades on day t.

We use the instrument vector including a constant, the change in the interest rate differential with a 1-month maturity, as the state vector to test the variance–covariance association between the changes in the exchange rate, the order flow and the bid–ask spread using the Wald test.[23] This setting implies that the number of moment conditions in Eqs. (10)–(14) is the same as the number of parameters, so the estimates come from the exact identification of the model.

4. Empirical results of liquidity and order flow effect

In this section, we address two key conclusions concerning the determinations of the FX rates. Consistent with Evans and Lyons (2002, 2008), our findings indicate that the order flow dominantly affects the FX rate return through the buying or selling pressures, but only the high trading density (active) currencies can support this conclusion. Our finding also indicates that the bid–ask spread has nontrivial influences on the FX rate returns for some currencies, especially when the explanatory power of the order flow is relatively less significant for the low trading density (infrequently traded) currencies.

In Table 2 we report the coefficients on the regressions of the FX rate returns on the order flow, the corresponding Newey-West

[22] We use the London inter-bank (overnight) offered rate supplied by the British Bankers' Association (BBA LIBOR) to compute the change in the interest rate differential in our empirical procedure (see Appendix B).

[23] Since Eqs. (10) and (11) are equivalent to the regression, $\Delta p_t = \gamma_0 + \gamma_1 \Delta x_t + \gamma_2 Spr_t + \varepsilon_t$ with two orthogonal conditions $E[\varepsilon_t \Delta x_t] = 0$ and $E[\varepsilon_t Spr_t] = 0$. We could take the variance on both sides of this regression to obtain the variance-covariance association between the second-order moments of the change in the exchange rate, order flow and Roll's bid-ask spread as follows: $Var_t(\Delta p_t) = \gamma_1^2 Var_t(\Delta x_t) + \gamma_2^2 Var_t(Spr_t) + 2\gamma_1\gamma_2 Cov_t(\Delta x_t, Spr_t) + Var_t(\varepsilon_t)$. We therefore use the Wald test to implement the following hypothesis tests: $\gamma_1^2 = 0, \gamma_2^2 = 0$ and $\gamma_1\gamma_2 = 0$.

Table B
List of interest rates corresponding to each currency.

Country/region	Currency name	Interest rate
Canada	Canadian Dollar	Canada interbank O/N offered rate (LIBOR)
Euro Zone	Euro Dollar	Euro interbank O/N offered rate (LIBOR)
Japan	Japanese Yen	Japan interbank S/N offered rate (LIBOR)
Mexico	Mexico Peso	Mexico banks acceptances 1 day – middle rate
New Zealand	New Zealand Dollar	New Zealand interbank S/N offered rate (LIBOR)
Norway	Norwegian Kroner	Norway interbank T/N offered rate (LIBOR)
Singapore	Singapore Dollar	Singapore interbank call – middle rate
Sweden	Swedish Krona	Sweden interbank S/N offered rate (LIBOR)
Switzerland	Swiss Franc	Switzerland interbank S/N offered rate (LIBOR)
United States	US Dollar	US interbank O/N offered rate (LIBOR)
United Kingdom	Pound Sterling	UK interbank O/N offered rate (LIBOR)

These abbreviations O/N, S/N and T/N stand for the maturities for which BBA LIBOR is fixed. There are 15 different maturities for each currency and day of fixing. The shortest maturity is overnight (O/N) for the Euro, US Dollar, Pound Sterling, and Canadian Dollar and spot/next (S/N) for all other currencies. An "overnight" (O/N) rate that you see quoted today will be valued today and mature tomorrow. A "spot/next" (S/N) ate that you see quoted today will be valued in 2 days (i.e., the day after tomorrow) and mature the day after that. A "tomorrow/next" (T/N) is used in deposit transactions and swaps when the first value date is tomorrow and maturity falls on the next working day.

Table C.1
Estimates of daily model for high trading density currencies.

Currency	Model 1					Model 2		
	Δx_{t-1}	*Adj* R^2	Spr_{t-1}	*Adj* R^2	$\Delta(i_{t-1} - i^*_{t-1})$	Δx_{t-1}	Spr_{t-1}	*Adj* R^2
The change in the spot exchange rate for the "given" price								
EUR/CHF	0.0176	−0.002	0.0008	−0.0031	0.0446	−0.0082	−0.472***	0.0111
	(0.52)		(1.38)		(1.04)	(−0.26)	(−3.13)	
EUR/GBP	0.1855*	0.0095	0.1240*	0.0172	0.0499	0.1918*	0.1263**	0.0252
	(1.79)		(2.02)		(0.50)	(1.83)	(1.98)	
EUR/JPY	−0.048	−0.002	0.0126	−0.0032	−0.1241	−0.0483	0.1046	−0.0068
	(−0.67)		(0.27)		(−0.67)	(−0.67)	(0.07)	
EUR/USD	−0.014	−0.001	1.137	−0.0021	−0.0334	−0.0151 (−0.80)	1.384	−0.0049
	(−0.75)		(0.65)		(−0.46)		(0.76)	
GBP/USD	−0.0082	−0.003	0.0195	−0.0006	0.1582**	−0.0119	0.0193	0.0086
	(−0.14)		(0.75)		(2.32)	(−0.21)	(0.72)	
USD/CAD	0.1916***	0.029	0.022***	0.0371	0.4719*	0.1161***	0.017***	0.0697
	(7.30)		(3.73)		(1.65)	(3.53)	(2.99)	
USD/CHF	−0.0834	0.003	−0.0035***	−0.003	0.0181	−0.0836	−0.004***	−0.0026
	(−1.46)		(−4.37)		(0.18)	(−1.46)	(0.42)	
USD/JPY	−0.0236	0.001	1.1474	−0.0026	−0.0697	−0.0235	1.0257	−0.0042
	(−1.19)		(0.32)		(−0.29)	(−1.16)	(0.28)	
The change in the spot exchange rate for the "paid" price								
EUR/CHF	0.0389	0.0032	0.001	−0.0031	0.0061	0.0183	−0.3808*	0.0095
	(1.17)		(1.60)		(0.08)	(0.62)	(−1.98)	
EUR/GBP	0.1214	0.0015	−0.3791***	0.1702	−0.0227	0.1193	−0.379***	0.1703
	(0.85)		(−3.52)		(−0.21)	(0.97)	(−3.48)	
EUR/JPY	−0.0445	−0.002	−0.0032	−0.0032	−0.1253	−0.0444	−0.0259	−0.0070
	(−0.62)		(−0.07)		(−0.68)	(−0.61)	(−0.02)	
EUR/USD	−0.0138	−0.001	1.3207	−0.0017	−0.0264	−0.0148	1.5461	−0.0049
	(−0.73)		(0.73)		(−0.36)	(−0.78)	(0.83)	
GBP/USD	0.0506	−0.0004	−0.0913**	0.0563	−0.128***	0.059	−0.092***	0.0640
	(1.07)		(−2.51)		(−2.93)	(1.22)	(−2.72)	
USD/CAD	0.1845***	0.0274	0.0176**	0.0226	0.257	0.133***	0.0115*	0.0375
	(8.43)		(2.46)		(1.31)	(4.27)	(1.77)	
USD/CHF	−0.0809	0.0032	−0.0038***	−0.0029	0.0269	−0.0812 (−1.43)	−0.004***	−0.0027
	(−1.43)		(−4.77)		(0.27)		(−4.70)	
USD/JPY	−0.0228 (−1.14)	0.0009	1.6124	−0.0020	−0.0895	−0.023	1.455	−0.0036
			(0.45)		(−0.37)	(−1.11)	(0.39)	

This table reports estimates of the daily regressions for high trading density currencies $\Delta p_t = \alpha_0 + \alpha_1 \Delta x_{t-1} + \omega_t$, $\Delta p_t = \beta_0 + \beta_1 Spr_{t-1} + \upsilon_t$ in Model 1 and $\Delta p_t = \delta_0 + \delta_1 \Delta(i_{t-1} - i^*_{t-1}) + \delta_2 \Delta x_{t-1} + \delta_3 Spr_{t-1} + \eta_t$ in Model 2 by using the Newey–West standard errors. The regressor $\Delta(i_t - i^*_t) = 100[(i_t - i^*_t) - (i_{t-1} - i^*_{t-1})]$ is the daily change in the interest rate differential between the local country and base country from day $t-1$ to day t. Descriptions of other variables are the same as those in Table 1. The Newey–West t-statistic is shown in the parentheses and *Adj* R^2 is the adjusted R-square used to detect the model's goodness of fit..

*** Significance at the 1% level.

** Significance at the 5% level.

* Significance at the 10% level.

t-statistic in the parentheses and the adjusted R-squares in Model 1. Clearly, this simple OLS estimation is sufficient to shed light on the paramount importance of the order flow for the active currencies regardless of the given or paid prices. Generally speaking, most of the regression coefficients on the order flows are positive and statistically significant for high trading density currencies at the 1% significance level.

Likewise, with reference to Model 2 in Table 2, the adjusted R-squares that measure the explanatory powers of the order flows range from a low of 7% for EUR/GBP to a high of 56.28% for USD/

Table C.2
Estimates of Daily Model for Low Trading Density Currencies.

Currency	Model 1					Model 2		
	Δx_{t-1}	*Adj* R^2	Spr_{t-1}	*Adj* R^2	$\Delta(i_{t-1} - i^*_{t-1})$	Δx_{t-1}	Spr_{t-1}	*Adj* R^2
The change in the spot exchange rate for the "given" price								
EUR/NOK	0.2097	−0.0037	−0.0031	−0.0037	−0.169***	0.5063	−0.0058	0.0012
	(0.25)		(−0.26)		(−3.20)	(0.64)	(−0.51)	
EUR/SEK	0.4255	−0.0016	−0.0611	−0.0014	0.1342	0.4053	−0.0873	−0.003
	(0.82)		(−0.87)		(1.16)	(0.80)	(−1.05)	
GBP/JPY	0.1134*	−0.0011	−0.0029	−0.0039	0.0299	0.1142*	−0.0018	−0.009
	(1.92)		(−0.08)		(0.11)	(1.92)	(−0.05)	
NZD/USD	1.3617*	−0.0024	0.2314***	0.0044	0.0688	0.7899	0.2212***	−0.003
	(1.77)		(3.96)		(0.07)	(0.88)	(3.68)	
USD/MXN	0.2888	−0.0038	0.6051***	0.1499	−0.219***	1.3763	0.6508***	0.1971
	(0.14)		(3.64)		(0.14)	(0.83)	(4.10)	
USD/SGD	0.0783	0.0545	0.6127***	0.3197	−0.035	0.0494	0.5855***	0.3382
	(1.21)		(6.95)		(−0.34)	(1.19)	(5.61)	
The change in the spot exchange rate for the "paid" price								
EUR/NOK	−3.0783*	0.0147	−0.1321***	0.1883	0.0274	0.5842	−0.1349***	0.1828
	(−2.03)		(−4.93)		(0.19)	(0.45)	(−4.90)	
EUR/SEK	0.2969	−0.0027	−0.4138**	0.1094	0.2444*	0.1392	−0.5537**	0.1544
	(0.67)		(−2.39)		(1.81)	(0.32)	(−2.34)	
GBP/JPY	−0.0231	−0.0037	−0.2642***	0.0687	−0.0719	−0.0311	−0.2637***	0.0618
	(−0.36)		(−5.50)		(−0.48)	(−0.61)	(−5.48)	
NZD/USD	0.1186	−0.0038	0.0496	−0.0012	0.2464	0.0047	0.0514	−0.008
	(0.18)		(1.42)		(0.47)	(0.01)	(1.39)	
USD/MXN	2.8666	0.0033	−0.3777***	0.0516	−0.124***	2.3399	−0.3463***	0.0642
	(1.49)		(−2.82)		(−3.74)	(1.24)	(−2.66)	
USD/SGD	0.0381	0.008	0.074	0.0002	−0.2506	0.0384	0.0481	0.0113
	(0.90)		(0.56)		(−1.21)	(0.99)	(0.39)	

*** Significance at the 1% level.
** Significance at the 5% level.
* Significance at the 10% level.

JPY. There are significantly positive associations between the order flow and exchange rate returns in most high trading density currency pairs, but EUR/GBP, GBP/USD and USD/CAD nevertheless have relatively low adjusted *R*-squares. There are low adjusted *R*-squares on EUR/GBP and GBP/USD perhaps because the Electronic Broking Service (EBS) now dominates in euro and yen instead of in sterling as discussed in Osler (2009).

In noticing the significant coefficients on the order flows (Δx_t), we are interested in the role of trading density in transmitting the information based on the order flow. For the three highest trading density currencies, EUR/USD, EUR/JPY and USD/JPY, the adjusted *R*-square is above 35%. These large adjusted *R*-squares suggest that the amount of information incorporated into the FX rates through the order flow increases as the level of the trading density increases.

The middle panel of Table 2 reports the coefficient estimates and adjusted *R*-squares of the realized bid–ask spread (Spr_t) in Model 1. At the 10% significance level, these coefficients are significant but with low *R*-squares except for EUR/USD, USD/CHF and USD/JPY. These results indicate that the spread may have the potential to determine the FX rate when the trading density declines. As a result, market quality depends not only on the level of trading activity, but also on the pattern of price reversal and price continuation in the transactions.

The right panel shows estimates for the specification of Model 2. It can obviously be seen that the coefficient on the order flow tends to be more significant than that on the bid–ask spread for the FX rate return. This is because more news about the pattern of price reversal and continuation are revealed by the buying or selling pressures. According to the above inspection, the results of Table 3 are totally different from the results of Table 2. They show that the coefficient on the bid–ask spread is relatively significant to the order flow for these infrequently traded currencies (e.g., GBP/JPY, NZD/USD, USD/MXN, and USD/SGD).

When comparing Table 2 with Table 3, we reach two conclusions. First, the order flow is more dominant than the spread in the determination of the FX rate return when the trading density is high. Second, the bid–ask spread has the potential explanatory power of the FX rate return except in the case of the three highest trading density currencies (e.g., EUR/USD, USD/CHF and USD/JPY). The empirical findings indicate that, instead of the order flow, the bid–ask spread has some informational content to explain the FX rate return in low trading density currencies. This contemporaneous association between exchange rate returns and the bid–ask spread supports the view that liquidity effects play an important role in determining the exchange rate when trading density is low.

A strong positive contemporaneous association between exchange rate returns and order flows has been documented in many recent empirical studies. However, Sager and Taylor (2008) and Cerrato et al. (2011) indicate that this contemporaneous order flow model does not work if it is replaced by the lagged order flow model. We also implement the lagged model to investigate whether this strong positive relationship is appropriate to the lagged order flow and the lagged realized bid–ask spread. The empirical results are reported in Appendix C.

Our finding is consistent with Sager and Taylor (2008) and Cerrato et al. (2011) and shows that the estimated coefficients and adjusted *R*-squared on the lagged variables have now declined drastically. The coefficient on lagged order flow is statistically insignificant in all currency pairs regardless of the high and low trading density.

This rules out a role for an order flow as a channel by which fundamental information is transmitted to the market. In a departure from Evans and Lyons (2002) and Evans and Lyons (2008), we show that the lagged order flow is unable to drive the change in the FX rate. As a result, the order flow is not working here any longer to reveal the fundamental macroeconomic information to the FX market.

Table D.1
Diagnostic tests for GMM estimates of daily high trading density currencies.

Currency	(i) Lagged order flow		(ii) Serial correlation						(iii) Residual ARCH						(iv) State-dependency					
			Δp_t		Δx_t		Spr_t		Δp_t		Δx_t		Spr_t		Δp_t		Δx_t		Spr_t	
	θ_1	θ_3	θ_1	θ_3	θ_1	θ_3	θ_1	θ_3	θ_1	θ_3	θ_1	θ_3	θ_1	θ_3	θ_1	θ_3	θ_1	θ_3	θ_1	θ_3
The change in the log spot exchange rate for the "given" price																				
EUR/CHF	0.05	0.00	0.44	0.29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.13	2.57	0.00	0.00	0.08	0.07	0.03	6.23**
	[0.83]	[0.97]	[0.51]	[0.59]	[1.00]	[1.00]	[1.00]	[1.00]	[0.98]	[0.99]	[0.96]	[0.99]	[0.71]	[0.11]	[0.98]	[0.96]	[0.78]	[0.79]	[0.86]	[0.02]
EUR/GBP	0.02	0.01	0.04	0.05	0.00	0.00	0.00	0.00	0.01	0.00	0.01	0.09	0.00	0.00	0.00	0.01	0.09	0.04	0.03	0.06
	[0.88]	[0.94]	[0.84]	[0.82]	[1.00]	[1.00]	[1.00]	[1.00]	[0.91]	[0.98]	[0.91]	[0.76]	[0.95]	[0.98]	[0.99]	[0.91]	[0.77]	[0.83]	[0.87]	[0.80]
EUR/JPY	0.01	0.00	0.18	0.07	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.25	0.02	0.00	0.01	0.02	0.00	1.05
	[0.75]	[0.97]	[0.67]	[0.79]	[1.00]	[1.00]	[1.00]	[1.00]	[0.96]	[0.91]	[0.99]	[0.99]	[0.99]	[0.61]	[0.87]	[0.99]	[0.94]	[0.89]	[0.97]	[0.31]
EUR/USD	0.00	0.07	0.03	0.08	0.00	0.00	0.00	0.00	0.00	0.01	0.09	0.03	0.02	0.22	0.03	0.00	0.05	0.00	0.02	0.53
	[0.95]	[0.79]	[0.86]	[0.78]	[1.00]	[1.00]	[1.00]	[1.00]	[0.95]	[0.93]	[0.76]	[0.86]	[0.89]	[0.64]	[0.87]	[0.99]	[0.82]	[0.98]	[0.89]	[0.47]
GBP/USD	0.14	0.04	0.21	0.04	0.00	0.00	0.00	0.00	0.00	0.09	0.00	0.12	0.03	0.08	0.00	0.00	0.00	0.00	0.03	0.06
	[0.71]	[0.84]	[0.65]	[0.84]	[1.00]	[1.00]	[1.00]	[1.00]	[0.99]	[0.76]	[0.98]	[0.73]	[0.86]	[0.78]	[0.99]	[0.99]	[0.97]	[0.98]	[0.87]	[0.80]
USD/CAD	0.10	0.05	2.72	0.44	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.14	0.31	0.00	0.00	0.00	0.00	0.18	1.06
	[0.75]	[0.83]	[0.10]	[0.51]	[1.00]	[1.00]	[1.00]	[1.00]	[0.96]	[0.96]	[0.98]	[0.97]	[0.29]	[0.58]	[0.95]	[0.96]	[0.99]	[0.99]	[0.67]	[0.30]
USD/CHF	0.13	0.01	0.02	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.09	0.00	0.00	0.02	0.01	0.00	0.03
	[0.72]	[0.91]	[0.89]	[0.79]	[1.00]	[1.00]	[1.00]	[1.00]	[0.99]	[0.95]	[0.94]	[0.96]	[0.99]	[0.77]	[0.96]	[0.96]	[0.88]	[0.91]	[0.98]	[0.86]
USD/JPY	0.25	0.24	0.27	0.02	0.02	0.00	0.00	0.00	0.00	0.03	0.00	0.01	0.00	0.00	0.00	0.00	0.01	0.12	0.00	0.01
	[0.62]	[0.62]	[0.60]	[0.89]	[0.89]	[1.00]	[1.00]	[1.00]	[0.95]	[0.87]	[0.95]	[0.94]	[0.99]	[0.98]	[0.97]	[0.99]	[0.91]	[0.72]	[0.98]	[0.91]
The change in the log spot exchange rate for the "paid" price																				
EUR/CHF	0.00	0.00	1.78	2.03	0.61	0.26	0.00	0.00	0.00	0.00	0.00	0.00	0.16	5.41**	0.00	0.00	0.03	0.03	0.03	4.27**
	[1.00]	[1.00]	[0.18]	[0.15]	[0.43]	[0.61]	[1.00]	[1.00]	[0.99]	[0.99]	[0.97]	[0.99]	[0.69]	[0.02]	[0.98]	[0.98]	[0.86]	[0.86]	[0.86]	[0.04]
EUR/GBP	0.00	0.00	0.12	0.37	0.02	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.00	0.00	0.00	0.01	0.00	0.01	0.03	0.00
	[1.00]	[1.00]	[0.73]	[0.54]	[0.89]	[0.94]	[1.00]	[1.00]	[0.99]	[0.99]	[0.93]	[0.91]	[0.96]	[0.99]	[0.95]	[0.94]	[0.99]	[0.92]	[0.86]	[0.97]
EUR/JPY	0.00	0.00	0.18	0.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.34	0.03	0.00	0.01	0.02	0.00	1.06
	[1.00]	[1.00]	[0.67]	[0.79]	[0.98]	[0.99]	[1.00]	[1.00]	[0.95]	[0.95]	[0.99]	[0.99]	[0.99]	[0.56]	[0.87]	[0.95]	[0.93]	[0.89]	[0.97]	[0.30]
EUR/USD	0.00	0.00	0.02	0.07	0.22	0.27	0.00	0.00	0.01	0.01	0.09	0.03	0.01	0.17	0.03	0.00	0.05	0.00	0.02	0.52
	[1.00]	[1.00]	[0.88]	[0.79]	[0.64]	[0.61]	[1.00]	[1.00]	[0.94]	[0.93]	[0.76]	[0.87]	[0.91]	[0.68]	[0.87]	[0.98]	[0.83]	[0.98]	[0.90]	[0.47]
GBP/USD	0.00	0.00	0.02	0.04	0.02	0.01	0.00	0.00	0.00	0.01	0.02	0.01	0.01	0.17	0.01	0.00	0.03	0.01	0.01	0.15
	[1.00]	[1.00]	[0.89]	[0.84]	[0.89]	[0.94]	[1.00]	[1.00]	[0.97]	[0.94]	[0.89]	[0.93]	[0.91]	[0.68]	[0.92]	[0.95]	[0.86]	[0.92]	[0.92]	[0.70]
USD/CAD	0.00	0.00	0.03	0.03	0.02	0.02	0.00	0.00	0.00	0.00	1.81	0.53	0.27	0.44	0.08	0.06	0.01	0.01	0.86	2.26
	[1.00]	[1.00]	[0.87]	[0.87]	[0.89]	[0.89]	[1.00]	[1.00]	[0.99]	[0.99]	[0.18]	[0.46]	[0.60]	[0.51]	[0.78]	[0.81]	[0.93]	[0.93]	[0.35]	[0.13]
USD/CHF	0.00	0.00	0.02	0.05	0.00	0.04	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.09	0.00	0.00	0.02	0.01	0.00	0.03
	[1.00]	[1.00]	[0.89]	[0.82]	[1.00]	[0.83]	[1.00]	[1.00]	[0.99]	[0.95]	[0.95]	[0.96]	[0.99]	[0.76]	[0.96]	[0.95]	[0.88]	[0.91]	[0.99]	[0.86]
USD/JPY	0.00	0.00	0.11	0.00	0.03	0.23	0.00	0.00	0.00	0.03	0.00	0.01	0.00	0.00	0.00	0.00	0.01	0.13	0.00	0.00
	[1.00]	[1.00]	[0.75]	[0.99]	[0.87]	[0.63]	[1.00]	[1.00]	[0.96]	[0.87]	[0.95]	[0.94]	[1.00]	[0.99]	[0.97]	[0.99]	[0.91]	[0.71]	[0.98]	[0.95]

The table only reports diagnostic Wald-statistics for some coefficients on the variance of the order flow, $\theta_1 = \lambda_1^2 - \gamma_1^2 = 0$ and the covariance of the order flow and bid–ask spread, $\theta_3 = \lambda_1\,\lambda_2 - \gamma_1\gamma_2 = 0$ (λ_1, γ_1 is the coefficient on the order flow and λ_2, γ_2 is the coefficient on the bid–ask spread for the diagnostic and original model, respectively). Coefficients are listed on the top with asymptotic *P*-values reported in square brackets based on the moment conditions in Eqs. (10)–(14) and the following conditions: (i) $E[(\Delta p_t - \lambda_1\Delta x_t - \lambda_2 Spr_t)\Delta x_{t-1}] = E[\varepsilon_t\Delta x_{t-1}] = 0$ for the lagged order flow; (ii) serial correlation: $E[(\Delta p_t - \lambda_1\Delta x_t - \lambda_2 Spr_t)(\Delta p_{t-1} - \lambda_1\Delta x_{t-1} - \lambda_2 Spr_{t-1})] = E[\varepsilon_t\varepsilon_{t-1}] = 0$, $E[\Delta x_t\Delta x_{t-1}] = 0$, $E[Spr_t Spr_{t-1}] = 0$ for Δp_t, Δx_t, Spr_t Eq., respectively; (iii) residual ARCH: $E[\{\Psi_t(\Delta p_t) - Var_t(\Delta p_t)\}\{\Psi_{t-1}(\Delta p_{t-1}) - Var_{t-1}(\Delta p_{t-1})\}] = 0$, $E[\{\Psi_t(\Delta x_t) - Var_t(\Delta x_t)\}\{\Psi_{t-1}(\Delta x_{t-1}) - Var_{t-1}(\Delta x_{t-1})\}] = 0$, $E[\{\Psi_t(Spr_t) - Var_t(Spr_t)\}\{\Psi_{t-1}(Spr_{t-1}) - Var_{t-1}(Spr_{t-1})\}] = 0$ for Δp_t, Δx_t, Spr_t Eq., respectively; (iv) The state-dependency test for daily trading density (n_t) has no effect on the residuals of the variances of the price changes, order flows and the bid–ask spreads, respectively: $E[\{\Psi_t(\Delta p_t) - Var_t(\Delta p_t)\}n_t] = 0$, $E[\{\Psi_t(\Delta x_t) - Var_t(\Delta x_t)\}n_t] = 0$ and $E[\{\Psi_t(Spr_t) - Var_t(Spr_t)\}n_t] = 0$. The asterisk $*$ denote significance at the 10% level..

** Significance at the 5% level.

Table D.2
Diagnostic tests for GMM estimates of daily low trading density pairs.

Currency	(i) Lagged order flow		(ii) Serial correlation						(iii) Residual ARCH						(iv) State-dependency					
			Δp_t		Δx_t		Spr_t		Δp_t		Δx_t		Spr_t		Δp_t		Δx_t		Spr_t	
	θ_1	θ_3	θ_1	θ_3	θ_1	θ_3	θ_1	θ_3	θ_1	θ_3	θ_1	θ_3	θ_1	θ_3	θ_1	θ_3	θ_1	θ_3	θ_1	θ_3
The change in the log spot exchange rate for the "given" price																				
EUR/NOK	0.27	0.18	0.30	0.00	0.00	0.00	0.00	0.00	0.27	0.11	0.44	0.00	0.27	0.04	0.01	0.31	0.17	0.78	0.05	0.52
	[0.60]	[0.67]	[0.58]	[0.98]	[1.00]	[1.00]	[1.00]	[1.00]	[0.61]	[0.74]	[0.51]	[0.98]	[0.60]	[0.84]	[0.90]	[0.58]	[0.68]	[0.38]	[0.83]	[0.47]
EUR/SEK	0.02	0.05	0.01	0.38	0.00	0.00	0.00	0.00	0.00	0.00	0.34	0.26	0.04	0.20	0.01	0.01	0.01	0.07	0.02	0.32
	[0.90]	[0.83]	[0.94]	[0.54]	[1.00]	[1.00]	[1.00]	[1.00]	[0.95]	[0.96]	[0.56]	[0.61]	[0.85]	[0.65]	[0.92]	[0.93]	[0.92]	[0.79]	[0.89]	[0.57]
GBP/JPY	0.00	0.00	0.02	0.02	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.09	0.11	0.00	0.00
	[0.94]	[0.95]	[0.90]	[0.89]	[1.00]	[1.00]	[1.00]	[1.00]	[0.99]	[0.93]	[0.97]	[0.96]	[0.99]	[1.00]	[0.99]	[0.99]	[0.76]	[0.73]	[0.99]	[0.99]
NZD/USD	0.22	0.05	0.11	0.01	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.01	0.00	0.09	0.02
	[0.64]	[0.83]	[0.74]	[0.90]	[1.00]	[1.00]	[1.00]	[1.00]	[0.93]	[0.99]	[0.97]	[0.98]	[0.91]	[0.96]	[0.98]	[0.99]	[0.94]	[0.96]	[0.76]	[0.88]
USD/MXN	0.03	0.00	0.04	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.00	0.00	0.02	0.00	0.02	0.03	0.04	0.11
	[0.86]	[0.99]	[0.84]	[0.86]	[1.00]	[1.00]	[1.00]	[1.00]	[0.95]	[0.97]	[0.93]	[0.92]	[1.00]	[1.00]	[0.89]	[0.95]	[0.88]	[0.87]	[0.83]	[0.73]
USD/SGD	0.29	0.01	0.07	0.12	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.03	0.00
	[0.59]	[0.92]	[0.79]	[0.72]	[1.00]	[1.00]	[1.00]	[1.00]	[0.95]	[0.94]	[0.93]	[0.96]	[1.00]	[1.00]	[0.96]	[0.95]	[0.93]	[0.93]	[0.87]	[0.97]
The change in the log spot exchange rate for the "paid" price																				
EUR/NOK	0.00	0.00	0.73	0.24	0.25	0.31	0.00	0.00	0.07	0.74	0.27	1.54	0.18	0.16	0.02	0.03	0.12	0.05	0.00	0.00
	[1.00]	[1.00]	[0.39]	[0.62]	[0.62]	[0.57]	[1.00]	[1.00]	[0.79]	[0.39]	[0.60]	[0.21]	[0.67]	[0.68]	[0.89]	[0.87]	[0.72]	[0.82]	[0.96]	[0.96]
EUR/SEK	0.00	0.00	0.05	0.07	0.01	0.00	0.00	0.00	0.00	0.00	0.02	0.06	0.05	0.03	0.00	0.00	0.02	0.01	0.00	0.00
	[1.00]	[1.00]	[0.82]	[0.78]	[0.94]	[0.94]	[1.00]	[1.00]	[0.96]	[0.98]	[0.89]	[0.81]	[0.82]	[0.87]	[0.96]	[0.99]	[0.90]	[0.93]	[0.99]	[0.99]
GBP/JPY	0.00	0.00	0.03	0.00	0.00	0.02	0.00	0.00	0.01	0.00	0.01	0.00	0.00	0.00	0.00	0.01	0.05	0.01	0.00	0.00
	[1.00]	[1.00]	[0.86]	[0.95]	[0.95]	[0.88]	[1.00]	[1.00]	[0.93]	[0.95]	[0.94]	[0.96]	[0.96]	[0.96]	[0.95]	[0.94]	[0.82]	[0.91]	[0.95]	[0.96]
NZD/USD	0.00	0.00	0.00	0.00	0.05	0.08	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00
	[1.00]	[1.00]	[0.97]	[0.96]	[0.82]	[0.77]	[1.00]	[1.00]	[0.98]	[0.99]	[0.98]	[0.99]	[0.96]	[0.99]	[0.99]	[0.99]	[0.96]	[0.97]	[0.94]	[0.99]
USD/MXN	0.00	0.00	0.04	0.14	0.00	0.00	0.00	0.00	0.03	0.00	0.01	0.01	0.00	0.00	0.00	0.00	0.01	0.01	0.02	0.03
	[1.00]	[1.00]	[0.84]	[0.71]	[0.96]	[1.00]	[1.00]	[1.00]	[0.86]	[0.98]	[0.92]	[0.93]	[0.99]	[0.99]	[0.97]	[0.96]	[0.93]	[0.94]	[0.90]	[0.87]
USD/SGD	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.02	0.03	0.09	0.00	0.00	0.00	0.01	0.02	0.04	0.11	0.32
	[1.00]	[1.00]	[0.99]	[0.99]	[1.00]	[1.00]	[1.00]	[1.00]	[0.91]	[0.88]	[0.86]	[0.77]	[1.00]	[1.00]	[0.95]	[0.93]	[0.89]	[0.84]	[0.74]	[0.57]

The asterisks *** denote significance at the 1% level, ** denote significance at the 5% level, and * denotes significance at the 10% level.

Similarly, the relationship between order flows and exchange rates can be stronger or weaker depending on whether or not the trading density in the lagged model is in line with the conclusion in the contemporaneous model as discussed in Bacchetta and van Wincoop (2006). The empirical results presented for this paper's lagged model seem to suggest instead that the micro-model in Evans and Lyons (2002) and Bacchetta and van Wincoop (2006) might be questionable when we are not allowed to connect the lagged order flow and bid–ask spread to the contemporaneous FX rate return.

We also discuss the cost structure of the bid–ask spread. To obtain the cost structure of the bid–ask spread, we first compute the hourly usual bid–ask spread $S_p = (P_a - P_b)/[2(P_a + P_b)]$, where P_a is the hourly FX rate for the ask (paid) price and P_b is the hourly FX rate for the bid (given) price. We then aggregate the hourly bid–ask spread figures with those for the daily usual bid–ask spread. Secondly, we compute the daily serial covariance of the change in the FX rate by taking the hourly covariance of the change in the FX rate and the next change in the FX rate for each currency pair to generate the series of the daily covariance of the change in the FX rate $Cov[\Delta R_t, \Delta R_{t+1}]$.

We then use Eq. (7) $Cov[\Delta R_t, \Delta R_{t+1}] = -\kappa S_p^2/4$ which is suggested by Glosten (1987) to obtain the daily cost structure of the bid–ask spread κ for each currency pair. For the purpose of comparison and to keep away from the small sample bias in the low trading density currency pair, we take the monthly average of the daily cost structure of the bid–ask spread for each currency pair and plot the series of the monthly cost structure of the bid–ask spread in Fig. 1.

Fig. 1 presents the proportion of the transaction costs that is due to adverse information for high and low trading density currencies. Those line plots indicate that, on average, the proportion of the traded spread not attributable to adverse-selection costs κ is 16.8% for high trading density currencies and 2.1% for low trading density currencies. As a consequence, the price impact on the low trading density currency is attributable to the adverse selection. Our estimates are in line with the results of Easley et al. (1996) in that the risk of information-based trading is lower for active securities than for infrequently traded securities. This is because the presence of relatively more uniformed traders in the low trading density currency reduces the probability that the liquidity supplier would end up trading with an informed trader.

Another interesting facet of our study concerns whether the daily information dispersion transmitted by the variance of the order flow or the deviation of the bid–ask spread will generate a significant disturbance on the variance of the FX rate return. We apply the standard two-step GMM estimates (without the serial correlation correction in the weighted matrix) to investigate the information dispersion and deviation of the price bounce. The relaxation of the GMM relative to the OLS is that it imposes very weak distributional assumptions.

In Table 4 we report parameter estimates on the order flow price-impact γ_1, the bid–ask spread price-impact γ_2, the expected variance of the price change $E[Var_t(\Delta p_t)]$, the expected variance of the order flow $E[Var_t(\Delta x_t)]$, and the expected variance of the change in the bid–ask spread $E[Var_t(Spr_t)]$ for high trading density currencies with exact identification. Asymptotic t-statistics allowing for residual serial correlation and heteroskedasticity are shown in the parentheses. It is obvious to see that the estimate of the order flow parameter γ_1 is positive and statistically significant, as the theory predicts, except for EUR/GBP, GBP/USD, and USD/CAD. The average magnitude of the significant γ_1 corresponding to a price impact roughly ranges from 20 to 72 basis points per $1 billion in the order flow.

Alternatively, at the 5% significance level, the coefficients on the bid–ask spread, γ_2 are significant except for EUR/CHF, USD/CHF and USD/JPY. These results indicate that the price impact of the order flow is comparatively larger than the price impact of the bid–ask spread for high trading density currencies. In addition, positive and significant variance parameters, $E[Var_t(\Delta p_t)]$, $E[Var_t(\Delta x_t)]$ and $E[Var_t(Spr_t)]$, show that expected variances of the change in the FX rate return, the order flow and the change in the spread are larger than zero.

To investigate whether the information dispersion or the deviation of the bid–ask spread could increase the volatility of the FX rate returns, we implement the Wald tests for the null that $\gamma_1^2 = 0, \gamma_2^2 = 0$ and $\gamma_1\gamma_2 = 0$ in the right panel of Table 4, in which γ_1^2 is highly significant at the 1% level for EUR/CHF, EUR/JPY, EUR/USD, USD/CHF and USD/JPY. Those results are symmetrically confirmed by the estimates of the "given" price reported in the upper panel and the "paid" price reported in the bottom panel of Table 4. Furthermore, most of the Wald statistics for the null that $\gamma_2^2 = 0$ and $\gamma_1\gamma_2 = 0$ are insignificant. Only the Wald statistics for EUR/GBP, GBP/USD and USD/CAD reject the null that $\gamma_2^2 = 0$, and the Wald statistics of EUR/JPY and EUR/USD reject the null that $\gamma_1\gamma_2 = 0$ at the 5% significance level. We therefore conclude that the volatility of the FX rate returns is attributable to the information dispersion for high trading density currencies.

As for the low trading density currencies, we find the opposite results in Table 5. Wald tests for the null that $\gamma_2^2 = 0$ tell us how the variance of the FX rate returns is impacted by the variance in the price reversal and continuation. This Wald statistic is positive and significant for EUR/NOK, GBP/JPY and USD/SGD at the 1% significance level. We present our robustness checks for the GMM in Appendix D. Once again, these specification tests in general reveal no evidence of significant misspecifications in our model.

5. Conclusion

This paper examines the implications of market microstructure for foreign exchange markets. We argue that the usual order flow model needs to be recast in broader terms to incorporate the transaction costs of liquidity and the limitation of price discovery through the order flow such as low trading density currencies. We measure the buying (selling) pressures by using the order flows and measure the liquidity by using the realized bid–ask spreads.

Using a daily data set with a distinct trading density, we find that the order flows have significant influences on the FX rate returns for high trading density currencies rather than for low trading density currencies. Alternatively, the bid–ask spreads also play nontrivial roles in the determination of the FX rates, especially for the low trading density currencies. This finding is evidenced by the positive association between the non-adverse selection components of the bid–ask spreads and the trading density. We therefore conclude that the risk associated with information-based trading is lower for high trading density currencies than for low trading density currencies.

We also implement the lagged model to investigate whether this strong positive relationship is appropriate to the lagged order flow and the lagged realized bid–ask spread. Consistent with Sager and Taylor (2008) and Cerrato et al. (2011), our results show that the estimated coefficients and adjusted R-squared value of the lagged variables have now declined drastically. The coefficient on the lagged order flow is statistically insignificant in all currency pairs regardless of the high and low trading density. These empirical results seem to suggest instead that the micro-model approach as in Evans and Lyons (2002) might be questionable when we are not allowed to connect the lagged order flow and bid–ask spread to the contemporaneous FX rate return.

Using the GMM model, we also find that the information dispersion of the FX rate return is greater when trading density is higher.

Overall, our work sheds light on the importance of the liquidity in the usual order flow model. Moreover, the information transmission of the order flow tends to be fruitless when the trading density is low.

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Appendix A. Derivation of serial covariance of transaction price changes

Based on the process of the transaction prices in Eqs. (4a) and (4b), Table A depicts the joint distribution of successive transaction price changes. In the parentheses next to each possible price change, $-\delta S$ or $(1-\delta)S$ is an indication of whether the transaction price is a continuation (BB or AA) or a reversal (BA or AB). Assuming that the last trade occurs with equal probability at the bid or ask price, the serial covariance of the transaction price change is:

$$\begin{aligned}\mathrm{cov}_T &\equiv Cov(\Delta R_t, \Delta R_{t+1}) \\ &= (1-\pi)^2\delta^2 S^2 - \delta S(1-\delta)S\pi(1-\pi) + \delta S(1-\delta)S\pi(1 \\ &\quad -\pi) - (1-\delta)^2 S^2\pi^2 \\ &= S^2[\delta^2(1-2\pi) - \pi^2(1-2\delta)] \end{aligned} \tag{A.1}$$

Appendix B

The London inter-bank (overnight) offered rate compiled by the British Bankers' Association (BBA LIBOR) is the most widely used "benchmark" or reference rate for short-term interest rates. It is compiled by the BBA in conjunction with Reuters and released to the market shortly after 11.00am London time each day. When the daily/overnight interest rate for a certain currency is not offered by BBA, we use the other interbank overnight rate or equivalence to replace it. All interest rate data are collected by Datastream and are listed in Table B.

Appendix C

Sager and Taylor (2008) highlight that regressions involving contemporaneous models may be spurious if a publication lag (i.e., lags of variables) is considered. We implement the following lagged model to investigate this important issue,

$$\Delta p_t = \alpha_0 + \alpha_1 \Delta x_{t-1} + \omega_t; \Delta p_t = \beta_0 + \beta_1 Spr_{t-1} + \upsilon_t, \tag{C.1}$$

$$\Delta p_t = \delta_0 + \delta_1 \Delta(i_{t-1} - i^*_{t-1}) + \delta_2 \Delta x_{t-1} + \delta_3 Spr_{t-1} + \eta_t. \tag{C.2}$$

By using the same EBS dataset representing most global interdealer transactions in the high trading density and low trading density currency pairs at the daily frequency over 2007, we find that a strong positive association is substantially weaker if contemporaneous independent variables are replaced by lagged variables (see Tables C.1 and C.2).

These contrasting results regarding the relationship between order flows and exchange rates have different implications concerning perfect foresight in the predictability of order flows, bid–ask spreads and interest rate differentials.

Appendix D

To provide additional robustness checks for these high and low trading density currencies, respectively, Table D.1 and D.2 present the results of diagnostic tests that examine the expanded set of moment conditions. In column (i), we report the Wald statistic for the difference between the original coefficient and the diagnostic coefficient on the variance of the order flow that $\theta_1 = \lambda_1^2 - \gamma_1^2 = 0$ using the orthogonal condition $E[(\Delta p_t - \lambda_1 \Delta x_t - \lambda_2 Spr_t)\Delta x_{t-1}] = E[\varepsilon_t \Delta x_{t-1}] = 0$ and moment conditions (10)–(14) for the lagged order flow, in which, λ_1, γ_1 is the coefficient on the order flow for the original GMM and diagnostic GMM, respectively.

Table D.1 and D.2 show that there is no significant evidence to reject this set of restrictions. Similarly, regarding the Wald statistic for the null that $\theta_3 = \lambda_1\lambda_2 - \gamma_1\gamma_2 = 0$, only the USD/JPY currency has significant differences between the original coefficients and the diagnostic coefficients on the covariance of the order flow and bid–ask spread, in which λ_2, γ_2 is the coefficient on the bid–ask spread for the original GMM and diagnostic GMM, respectively. The statistics in column (ii) test for the presence of (residual) serial correlation in the price changes, order flow and bid–ask spread process by, respectively, adding $E[\varepsilon_t\varepsilon_{t-1}] = 0$, $E[\Delta x_t \Delta x_{t-1}] = 0$, $E[Spr_t Spr_{t-1}] = 0$ to the original conditions (10)–(14). Again, as the results in column (ii) show, the Wald statistics for the coefficient differences in the variance of the order flow and the covariance of the order flow and bid–ask spread are insignificant. These results are consistent with the assumed structure of our model. Finally, in column (iii), we turn to the issue of state-dependency. Our daily model assumes that daily trading density (an accumulated number of intra-daily trades, n_t) has no effect on the residuals of the variances of the price changes, the residuals of the variances of the order flows or the residuals of the variances of the bid–ask spread by, respectively, adding $E[\{\Psi_t(\Delta p_t) - Var_t(\Delta p_t)\}n_t] = 0$, $E[\{\Psi_t(\Delta x_t) - Var_t(\Delta x_t)\}n_t] = 0$, and $E[\{\Psi_t(\Delta Spr_t) - Var_t(Spr_t)\}n_t] = 0$ to the original conditions (10)–(14). As the table shows, neither of the Wald statistics is significant.

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